

ML

Introduction to Logistic Regression

Logistic Regression

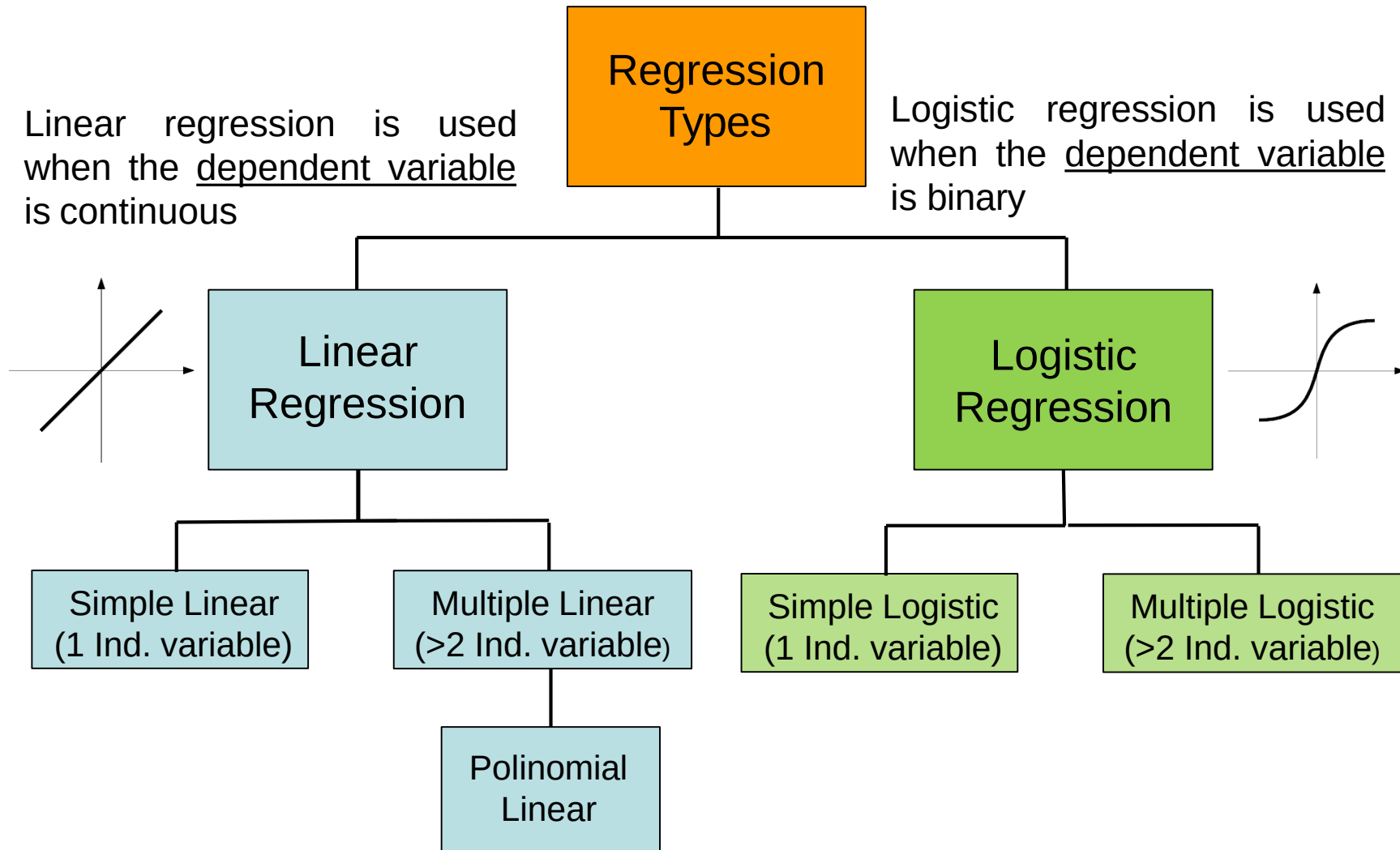
- **Definition**

Logistic Regression is a type of regression analysis used to predict a dependent variable (outcome variable) which can adopt a limited number of categories depending on the independent variables.

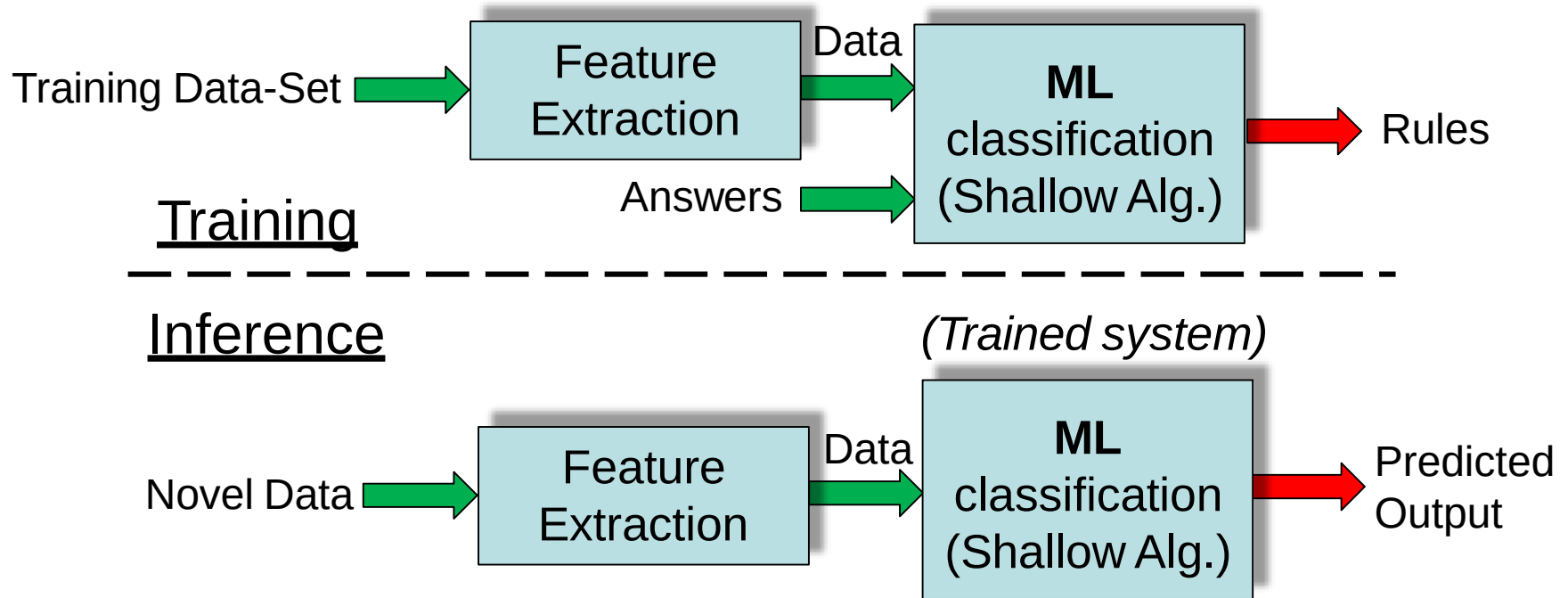
Machine Learning

- Supervised learning
 - Linear Regression
 - **Logistic Regression**
 - Unsupervised
 - Clustering
 - Market basket
 - Dimensionality Reduction
 - Reinforcement & Semi-supervised
- Logistic Regression
 - K-Nearest Neighbors (K-NN)
 - SVC
 - Naïve Bayes
 - Decision Tree – Classification
 - Random Forests - Classification

Regression Types

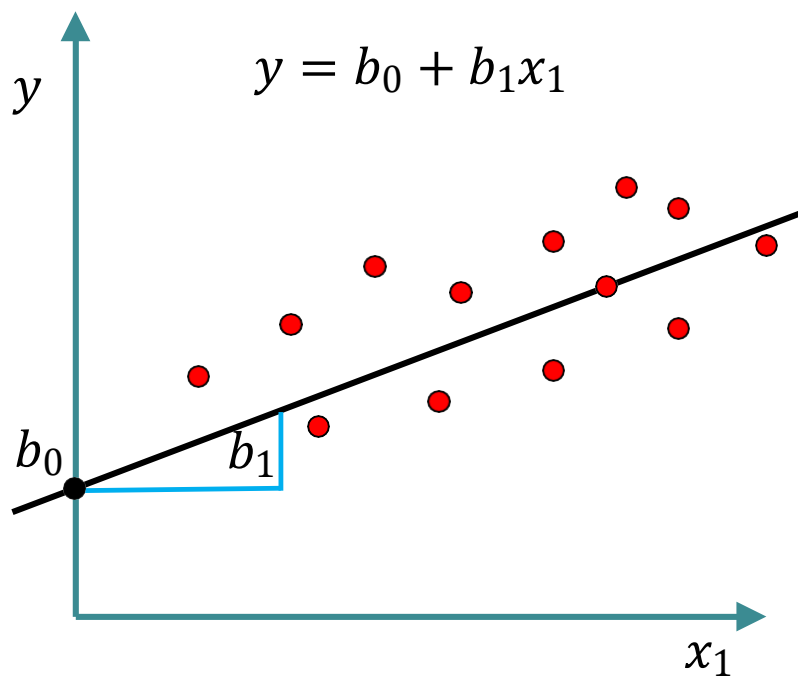


Machine Learning – Supervised Learning Approach

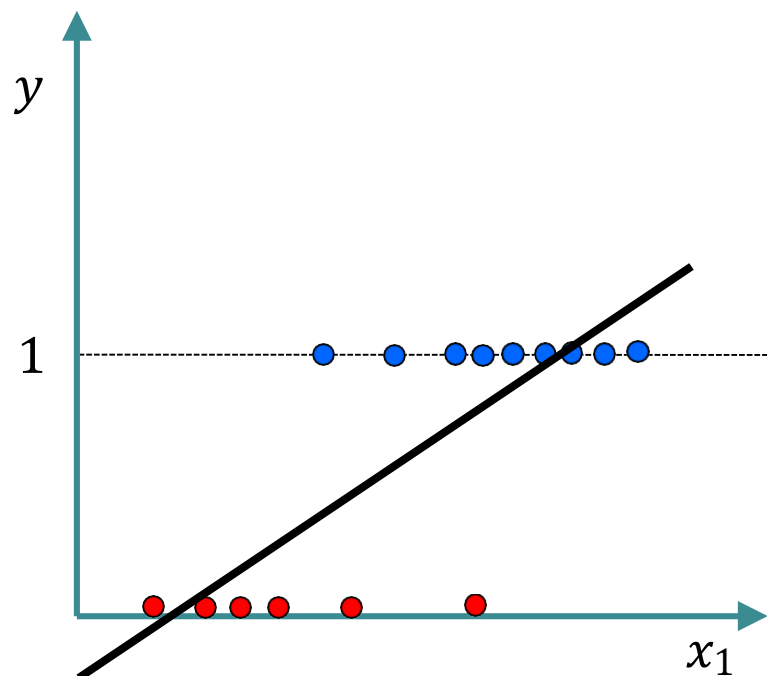


Logistic Regression (intuitive idea)

Lineal Regression

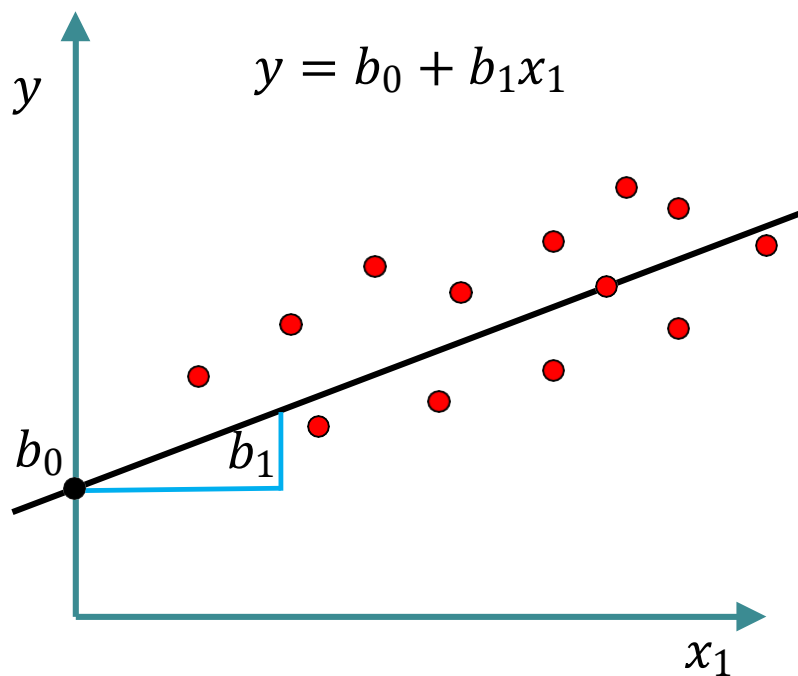


Logistic Regression

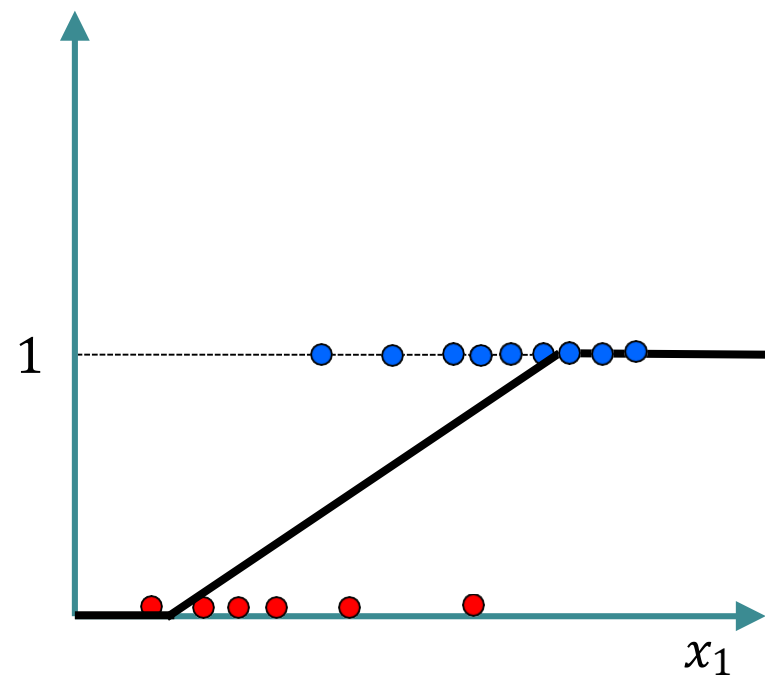


Logistic Regression (intuitive idea)

Lineal Regression



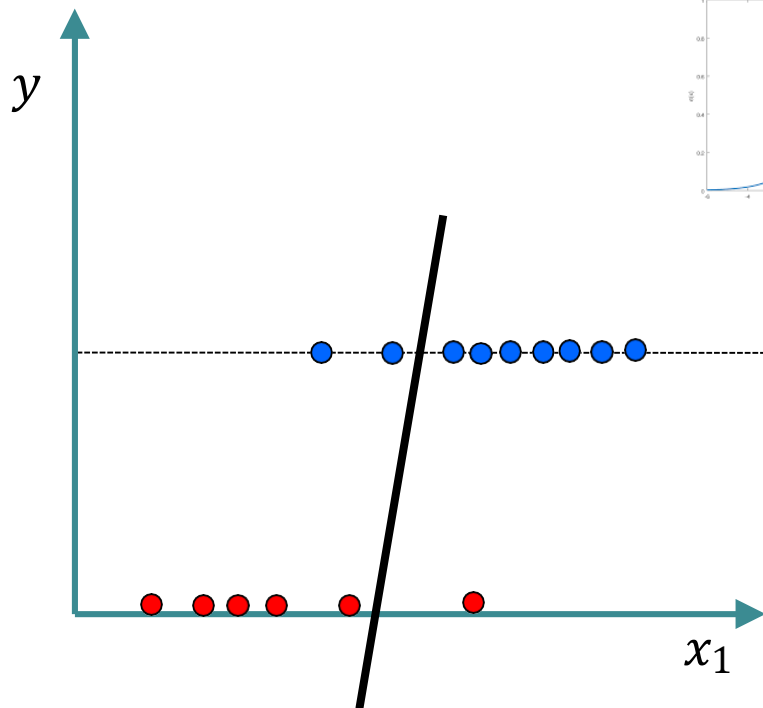
Logistic Regression



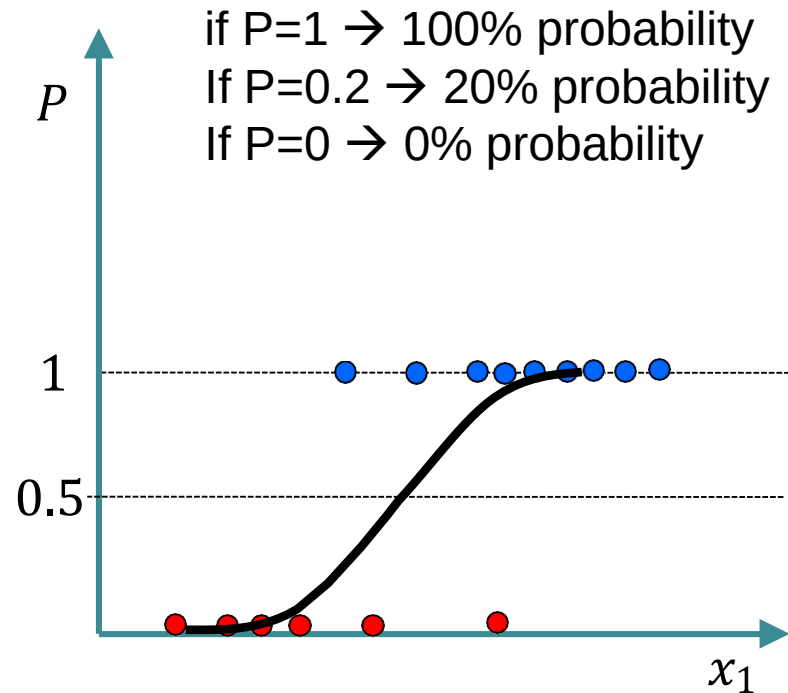
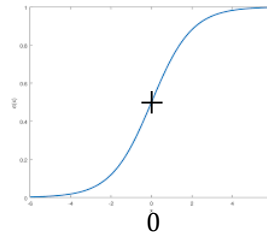
Logistic Regression (intuitive idea)

Sygmoid function $\longrightarrow$ logistic function

$$y = b_0 + b_1 x_1 \quad \longrightarrow \quad f(y) = \frac{1}{1 + e^{-y}} \quad \longrightarrow \quad P = f(y)$$



Lineal Regression

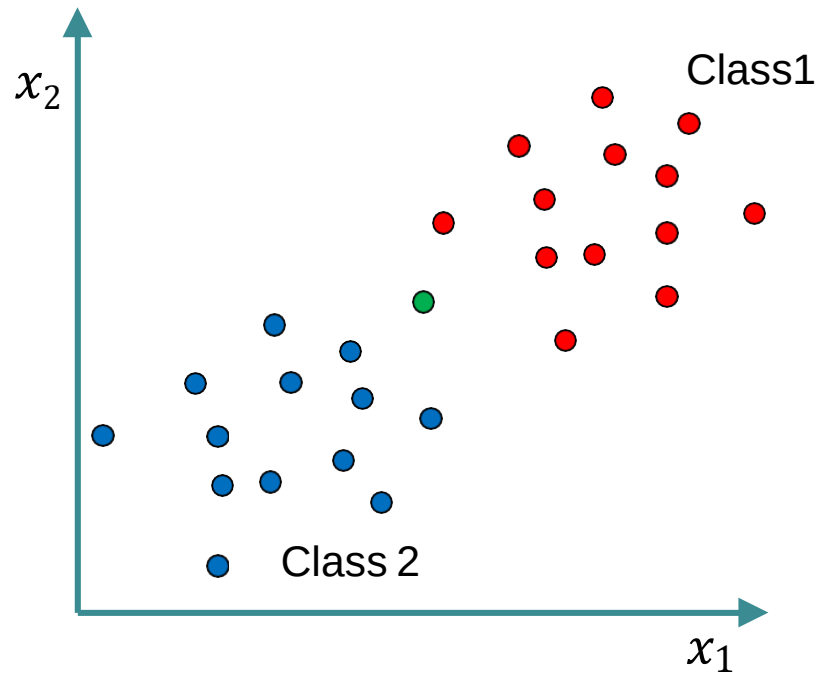


if $P=1 \rightarrow 100\%$ probability
If $P=0.2 \rightarrow 20\%$ probability
If $P=0 \rightarrow 0\%$ probability

Logistic Regression

Introduction to K-Nearest Neighbors (K-NN)

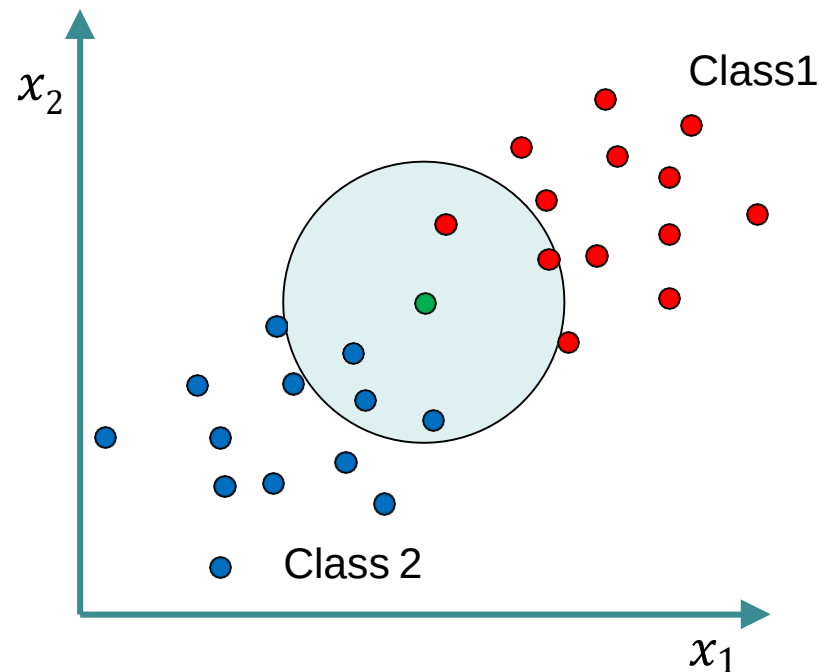
K-Nearest Neighbors (intuitive idea)



K-Nearest Neighbors (intuitive idea)

Algorithm

- 1º Set the k-number of the neighbor
- 2º Select the k-nearest data to the new data
- 3º Among the selected K-neighbors, count the number of points belonging to each class
- 4º Assign the new data to the class with the most neighbors



Assume:

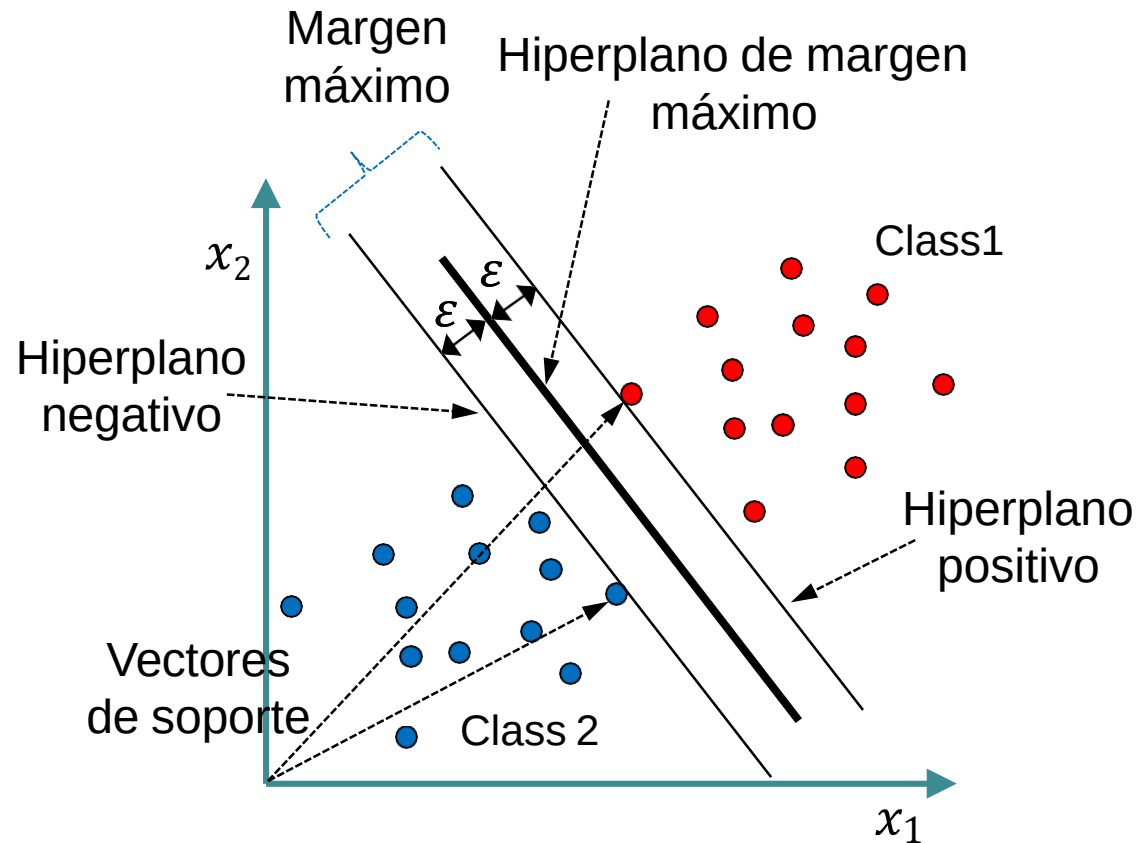
- K=5
- Euclidean distance



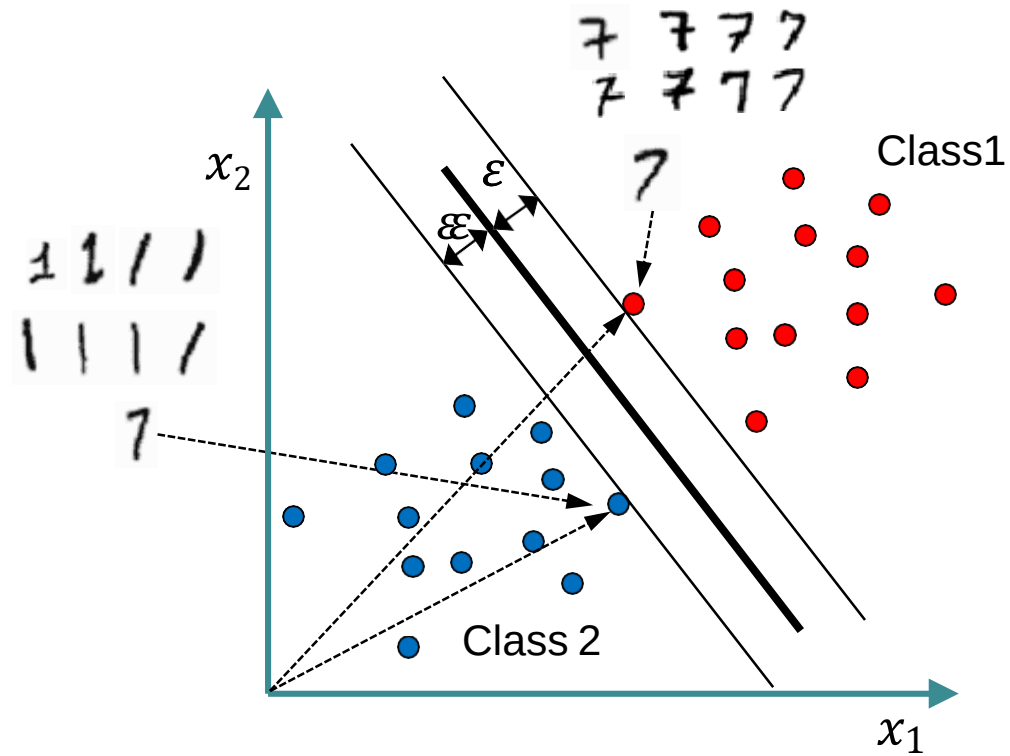
The new point belongs to Class 2

SVM for Classification (SVC)

SVC (kernel linear)

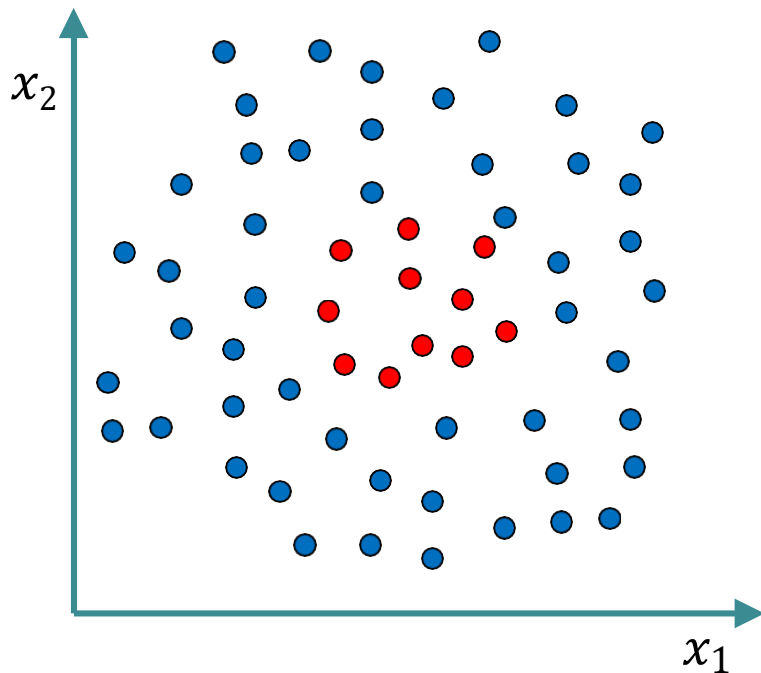


SVC (kernel linear)

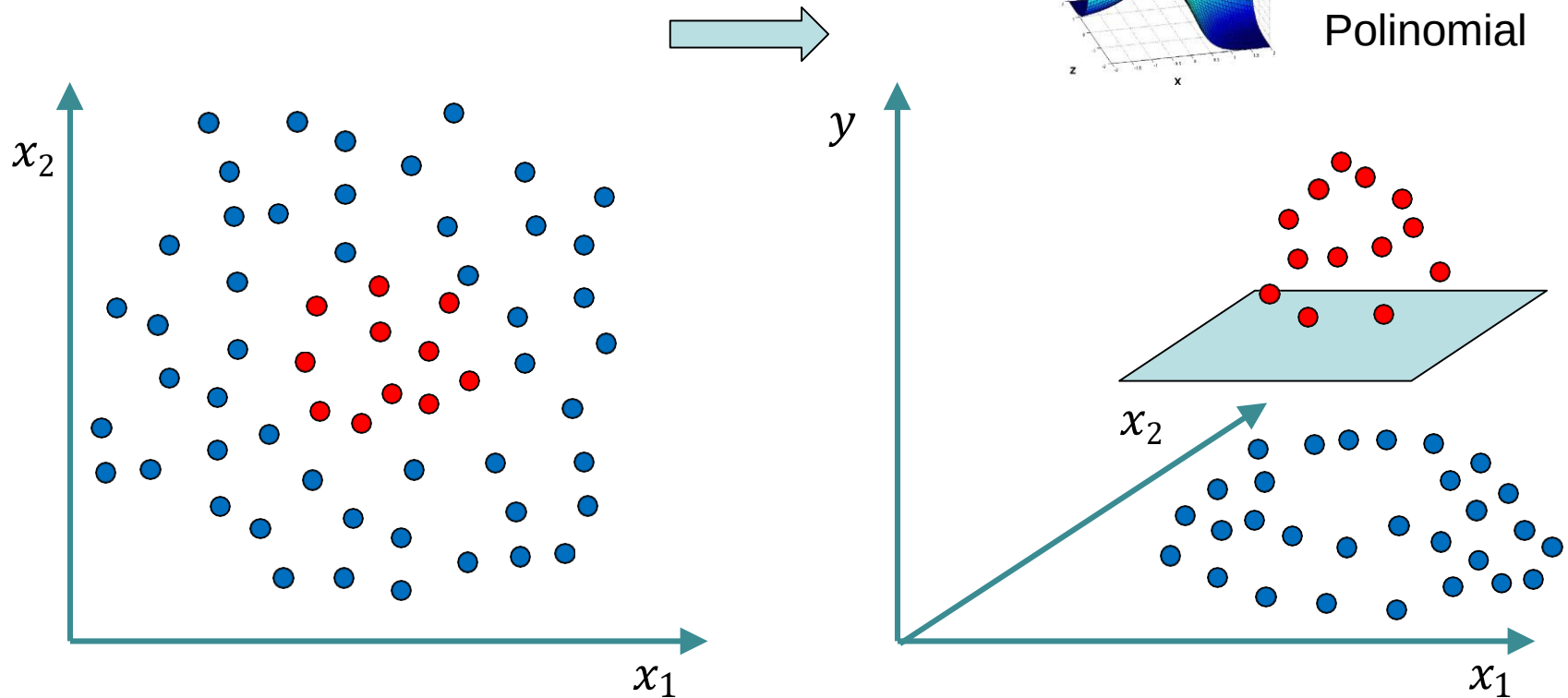


SVC (kernel no linear)

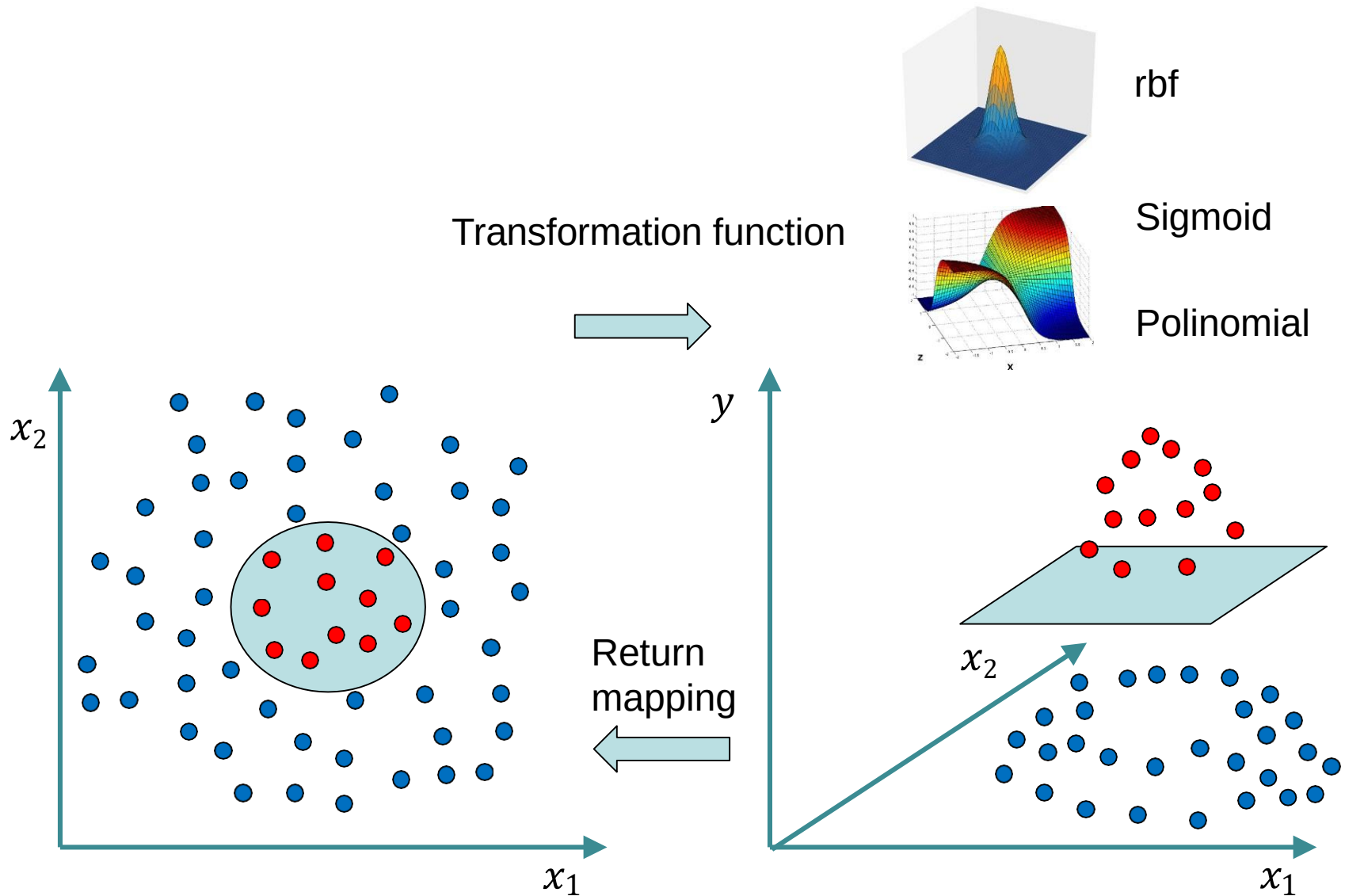
What if a problem is not linearly separable?



SVC (kernel no linear)



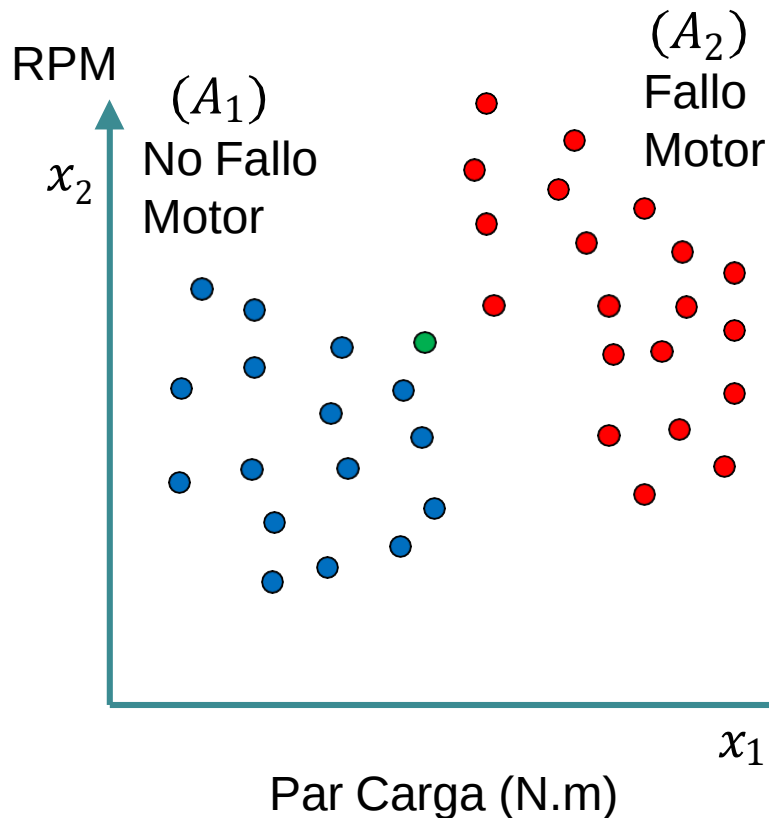
SVC (kernel no linear)



Naïve Bayes

Naïve Bayes

$$P(A_i|X) = \frac{P(X|A_i) \cdot P(A_i)}{P(X)}$$



A_i = Categoría de un suceso(No Fallo/ Fallo)

X = Características (Antigüedad / RPM)

|= Sabiendo que se cumple ...

$P(A_i|X)$ = Probabilidad a Posteriori

$P(A_i)$ = Probabilidad a Priori

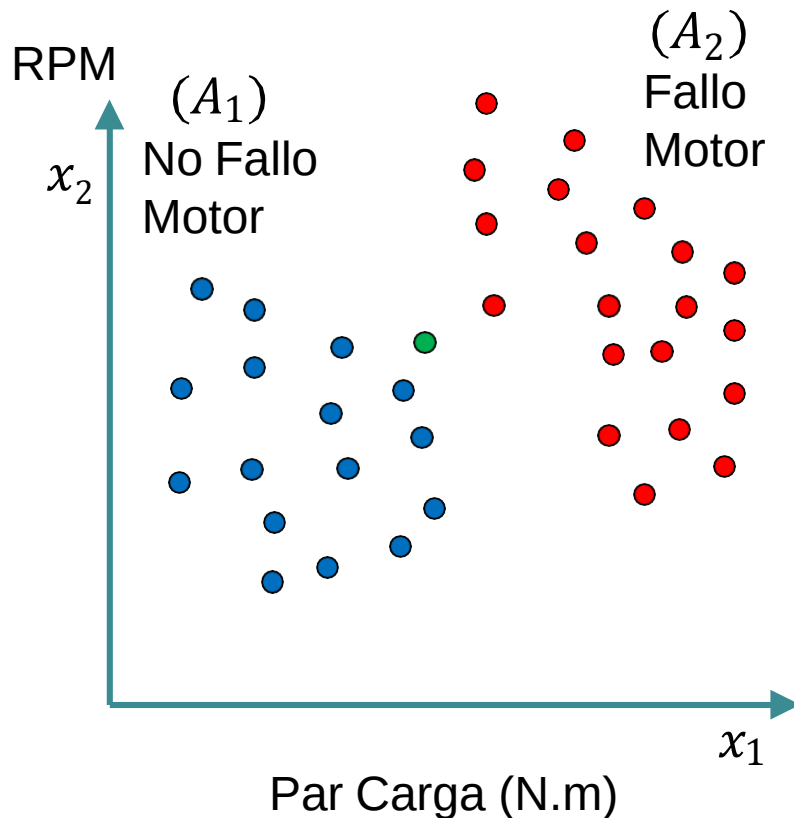
$P(X)$ = Probabilidad Marginal

$P(X|A_i)$ = Probabilidad Condicionada

$P(A_1|X)$ vs $P(A_2|X)$

Naïve Bayes

$$P(A_i|X) = \frac{P(X|A_i) \cdot P(A_i)}{P(X)}$$



A_i = Categoría de un suceso(No Fallo/ Fallo)

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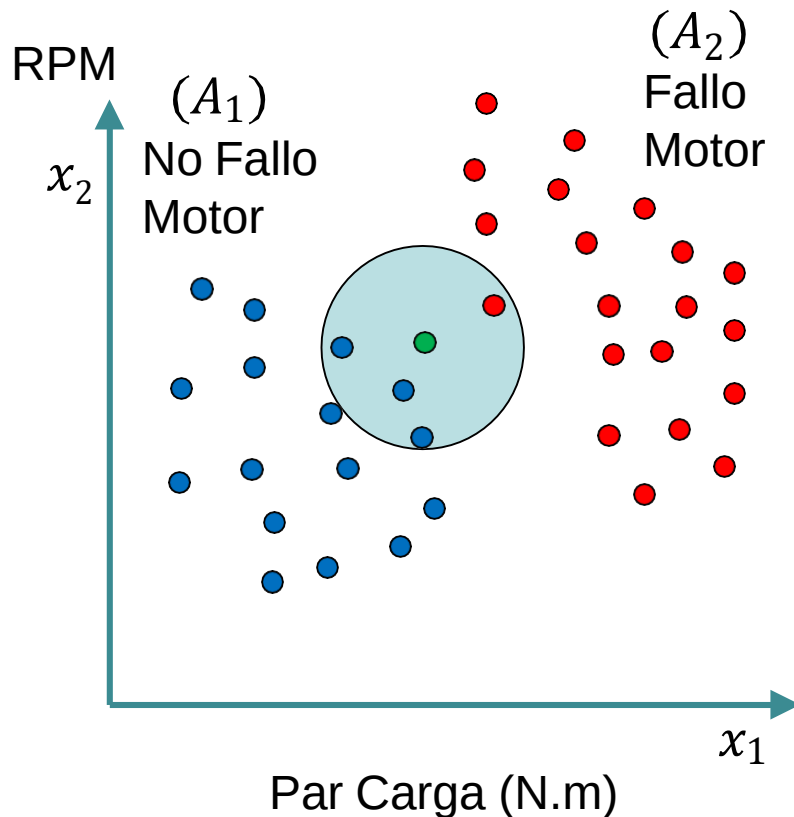
$P(A_1|X)$ vs $P(A_2|X)$

$$P(A_1) = P(\text{No Fallo}) = \frac{\text{Nº of No Fallos}}{\text{Tot. Observations}} = \frac{16}{36}$$

$$P(X) =$$

Naïve Bayes

$$P(A_i|X) = \frac{P(X|A_i) \cdot P(A_i)}{P(X)}$$



A_i = Categoría de un suceso(No Fallo/ Fallo)

X = Características (Par Carga / RPM)

$|$ = Sabiendo que se cumple ...

$P(A_i|X)$ = Probabilidad a Posteriori

$P(A_i)$ = Probabilidad a Priori

$P(X)$ = Probabilidad Marginal

$P(X|A_i)$ = Probabilidad Condicionada

$P(A_1|X)$ vs $P(A_2|X)$

$$P(A_1) = P(\text{No Fallo}) = \frac{\text{Nº of No Fallos}}{\text{Tot. Observations}} = \frac{16}{36}$$

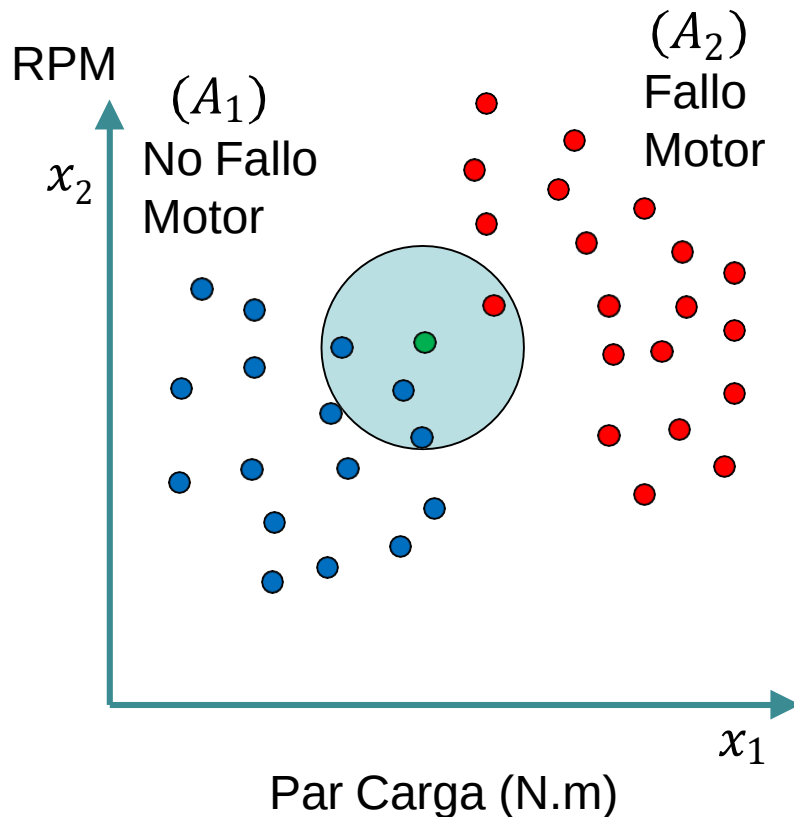
$$P(X) = \frac{\text{Nº of Similar Observations}}{\text{Tot. Observations}} = \frac{4}{36}$$

$$P(X|A_1) = \frac{\text{Nº Similar Obs. with No Fallos}}{\text{Nº Observation of No Fallos}} = \frac{3}{16}$$

$$P(A_1|X) = \frac{\frac{3}{16} \cdot \frac{16}{36}}{\frac{4}{36}} = 0.75$$

Naïve Bayes

$$P(A_i|X) = \frac{P(X|A_i) \cdot P(A_i)}{P(X)}$$



A_i = Categoría de un suceso(No Fallo/ Fallo)

X = Características (Antigüedad / RPM)

$|$ = Sabiendo que se cumple ...

$P(A_i|X)$ = Probabilidad a Posteriori

$P(A_i)$ = Probabilidad a Priori

$P(X)$ = Probabilidad Marginal

$P(X|A_i)$ = Probabilidad Condicionada

$P(A_1|X)$ vs $P(A_2|X)$

$$P(A_2) = P(\text{Fallo}) = \frac{\text{Nº of Fallos}}{\text{Tot. Observations}} = \frac{20}{36}$$

$$P(X) = \frac{\text{Nº of Similar Observations}}{\text{Tot. Observations}} = \frac{4}{36}$$

$$P(X|A_2) = \frac{\text{Nº Similar Obs. with Fallos}}{\text{Nº Observation of Fallos}} = \frac{1}{20}$$

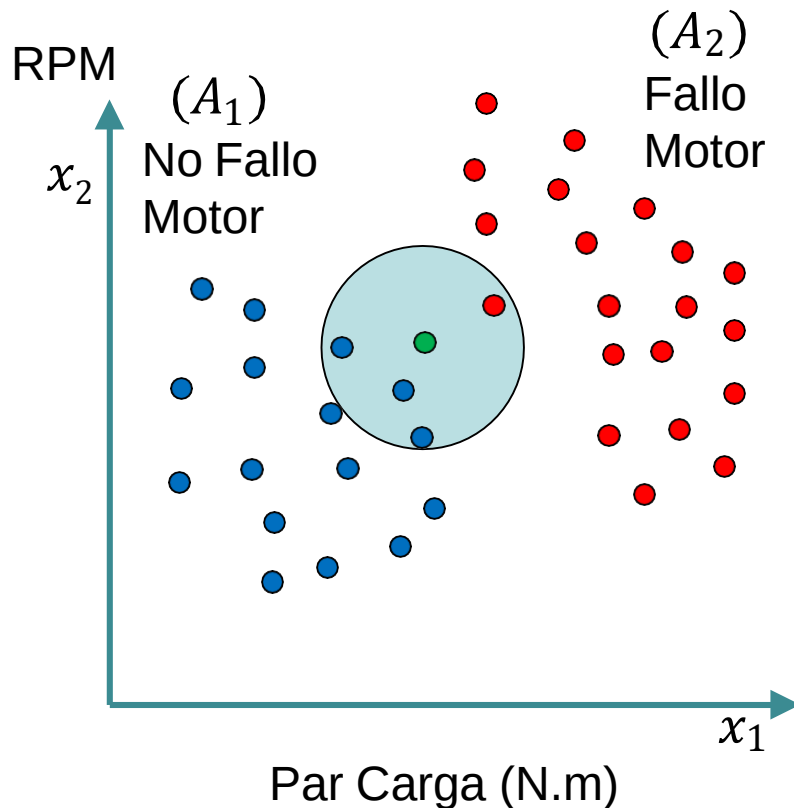
$$P(A_2|X) = \frac{\frac{1}{20} \cdot \frac{20}{36}}{\frac{4}{36}} = 0.25$$

Naïve Bayes

$$P(A_i|X) = \frac{P(X|A_i) \cdot P(A_i)}{P(X)}$$

$$P(A_1|X) = \frac{\frac{3}{16} \cdot \frac{16}{36}}{\frac{4}{36}} = 0.75$$

$$P(A_2|X) = \frac{\frac{1}{20} \cdot \frac{20}{36}}{\frac{4}{36}} = 0.25$$



$P(A_1|X) \text{ vs } P(A_2|X) \rightarrow \text{Class } A_1$

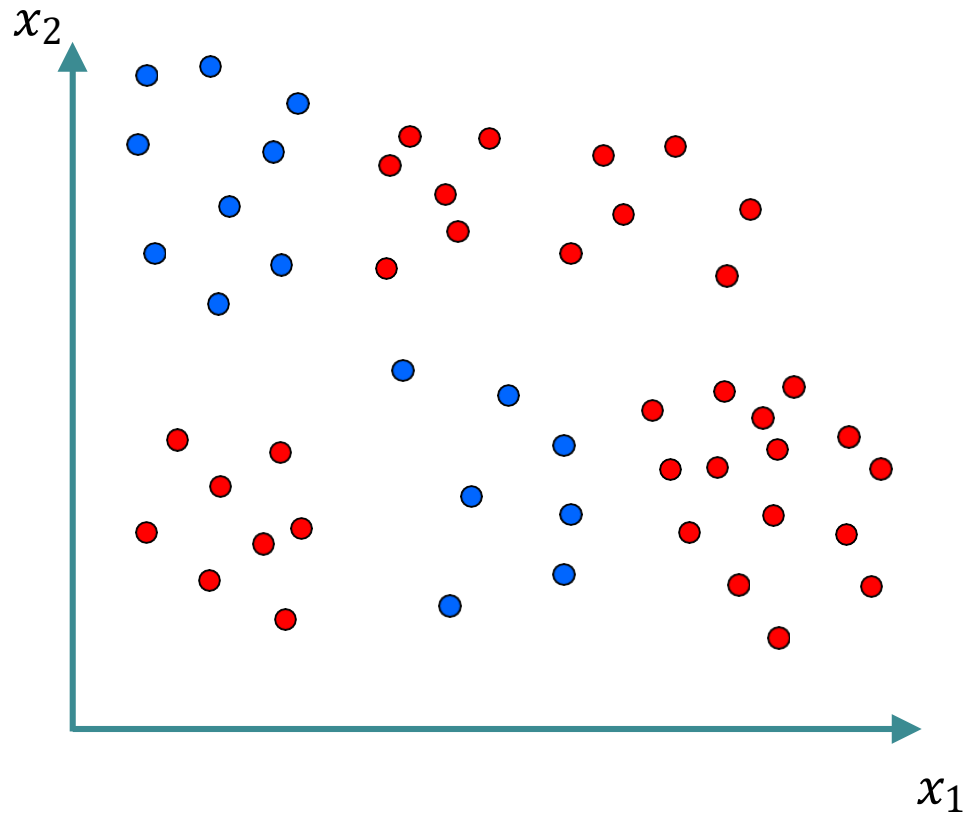
☐ *Naïve* ?

☐ $P(X)$?

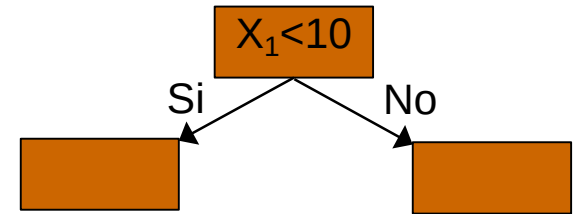
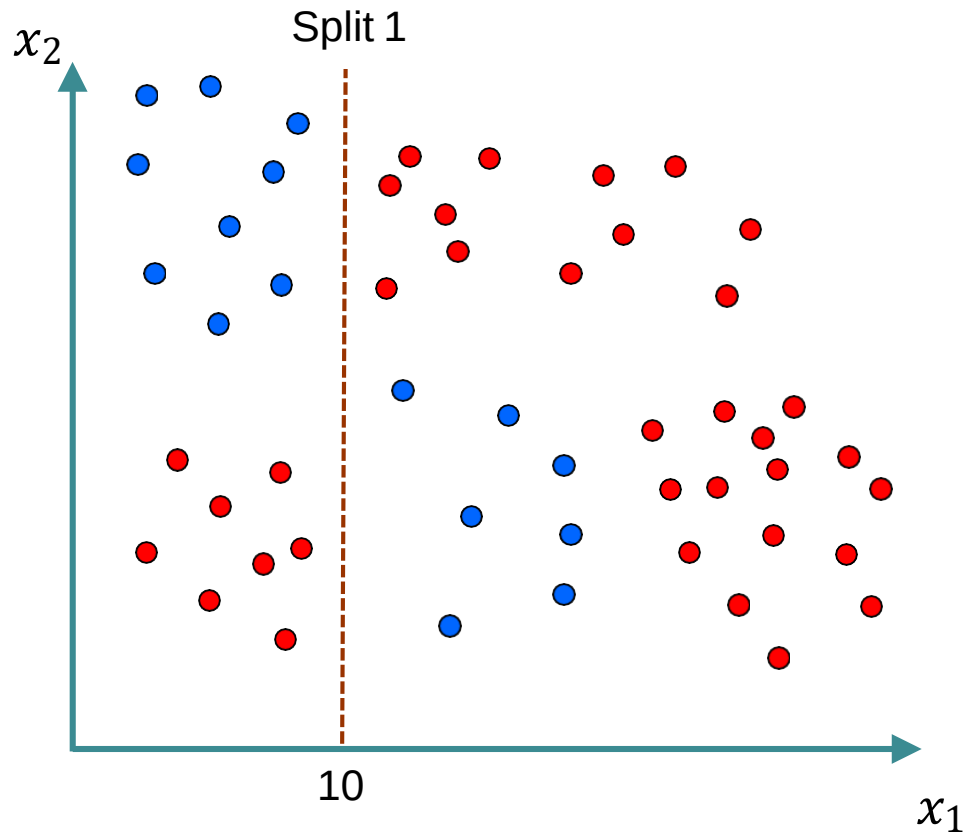
☐ What If More than 2 Class?

Decision Tree for Classification

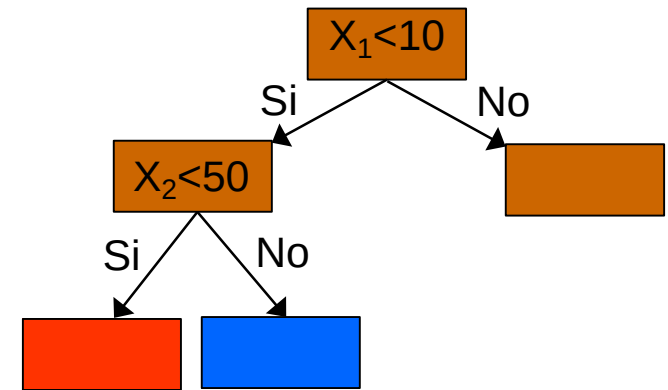
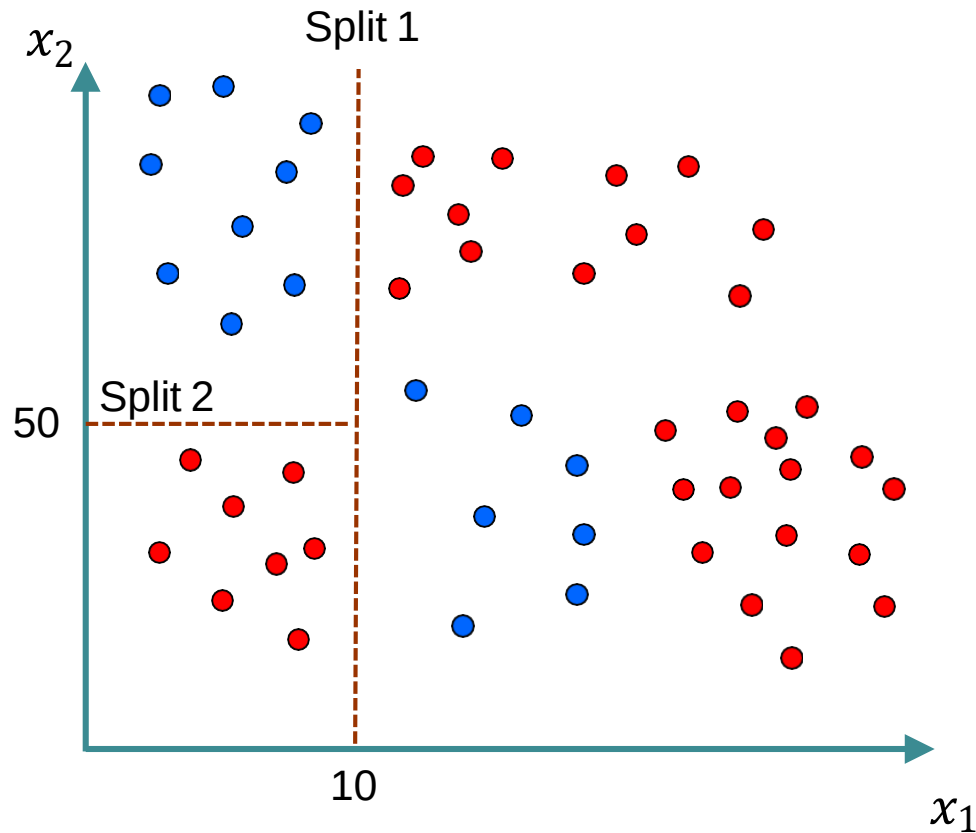
Decision Tree - Regression



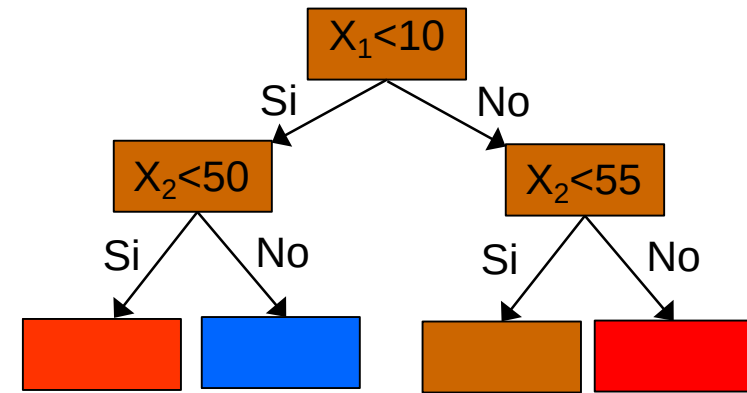
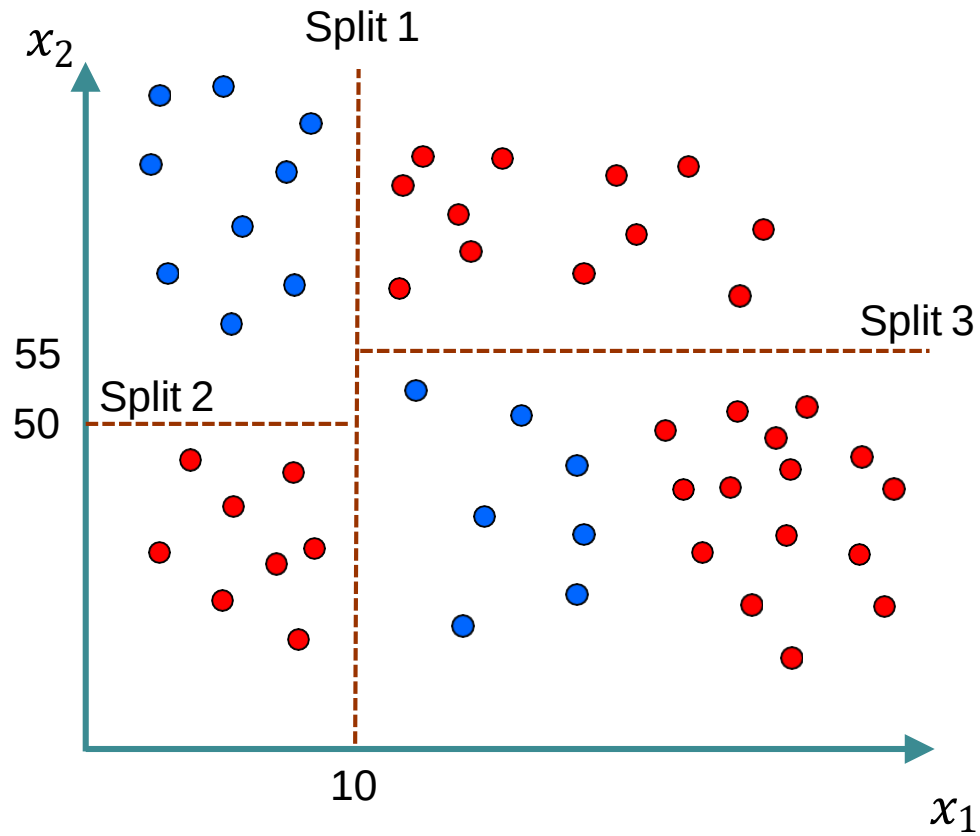
Decision Tree - Regression



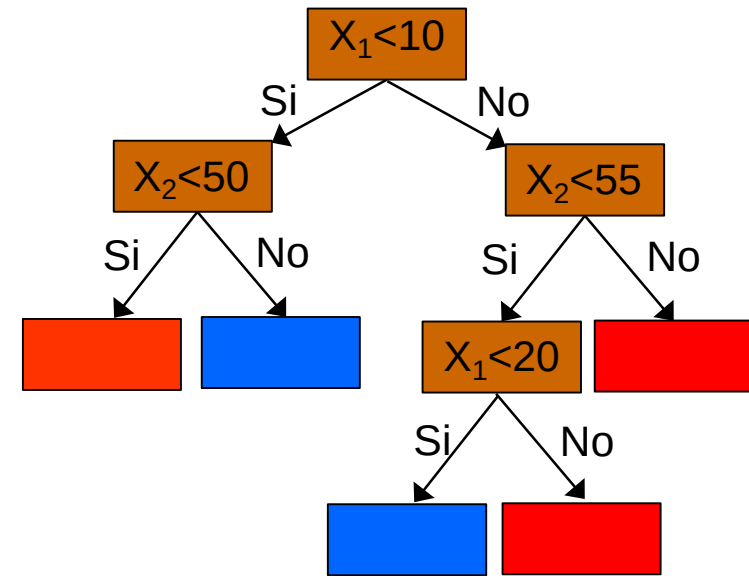
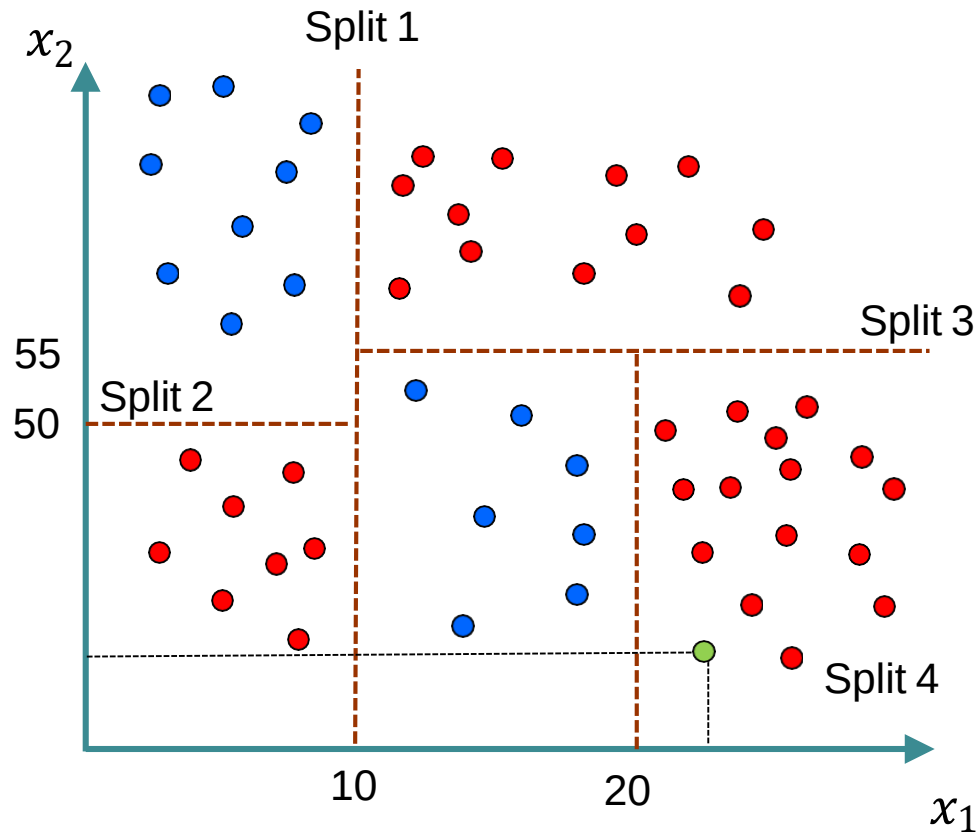
Decision Tree - Regression



Decision Tree - Regression



Decision Tree - Regression



$$P(x_1, x_2) = P(22, 7) \rightarrow \text{Red}$$

Random forests for Classification

Random Forests

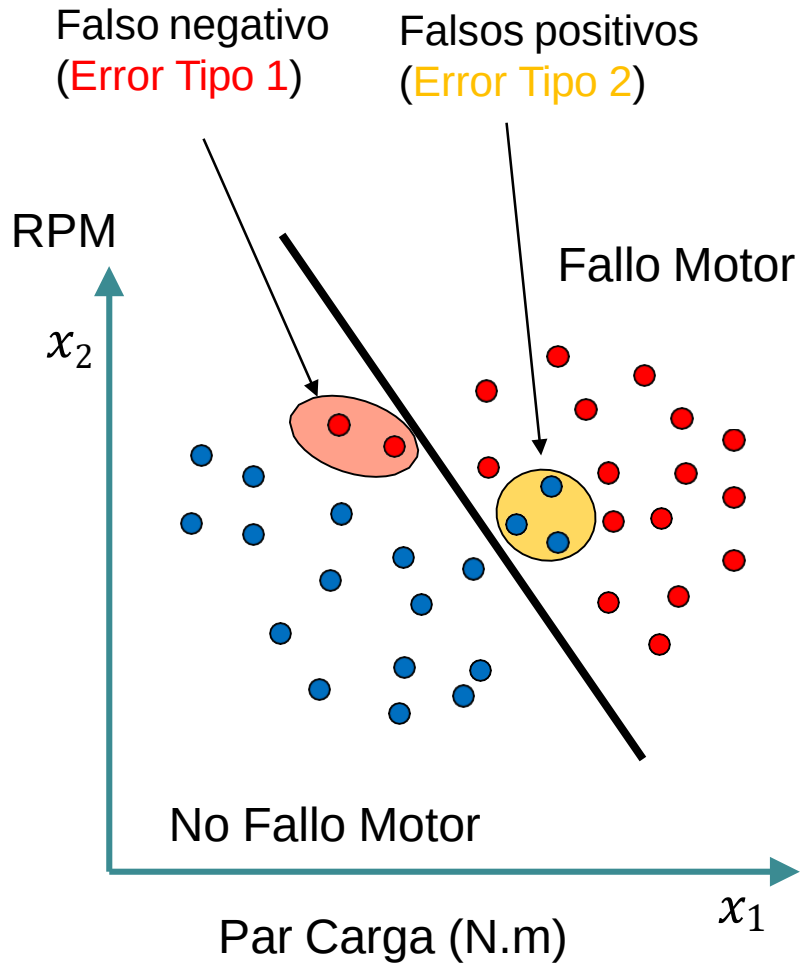
Stages

- 1° Set a random number M of data points from the training set**
- 2° Create a Decision Tree associated with those M points**
- 3° Set the number of Decision Trees to build and repeat steps 1 and 2.**
- 4° For a new test point make each of the n-trees predict a category**
- 5° Assign to the new test point the most voted category.**

<https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/BodyPartRecognition.pdf>

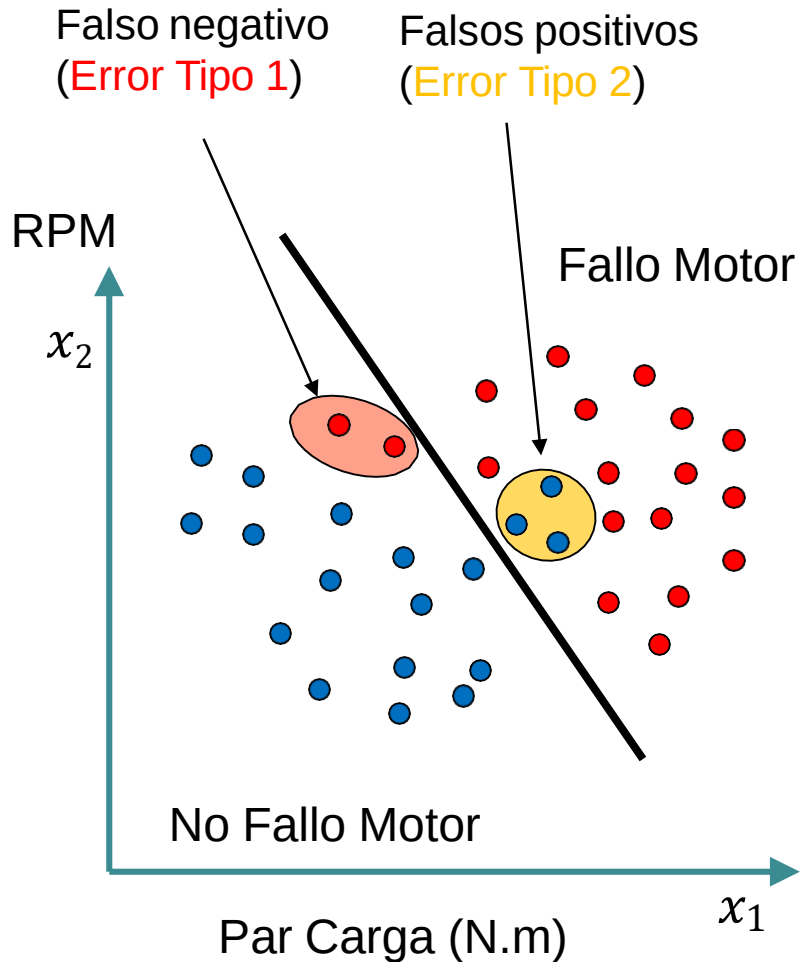
Classification, Models Evaluation

Confusion Matrix



		Resultado Predicho		
		Fallo M (0)	Fallo M (1)	
Resultado Real	True Neg. (TN)			
	Fallo M (0)	15	3	False Pos (FP) Error Tipo 2
	Fallo M (1)	2	16	True Pos (TP)
	False Neg (FN) Error Tipo 1			

Confusion Matrix



		<u>Resultado Predicho</u>			
		True Neg. (TN)	Fallo M (0)	Fallo M (1)	
<u>Resultado Real</u>	Fallo M (0)	15	3	False Pos (FP) Error Tipo 2	
	Fallo M (1)	2	16	True Pos (TP)	
		False Neg (FN) Error Tipo 1			
METRÍCS					

METRÍCS

$$\text{Accuracy: } ACC = \frac{TP+TN}{TP+TN+FP+FN} = \frac{31}{36} = 0.86$$

$$\text{Precision: } PPV = \frac{TP}{TP+FP} = \frac{16}{19} = 0.84$$

$$\text{Sensitivity: } TPR = \frac{TP}{TP+FN} = \frac{16}{18} = 0.88$$

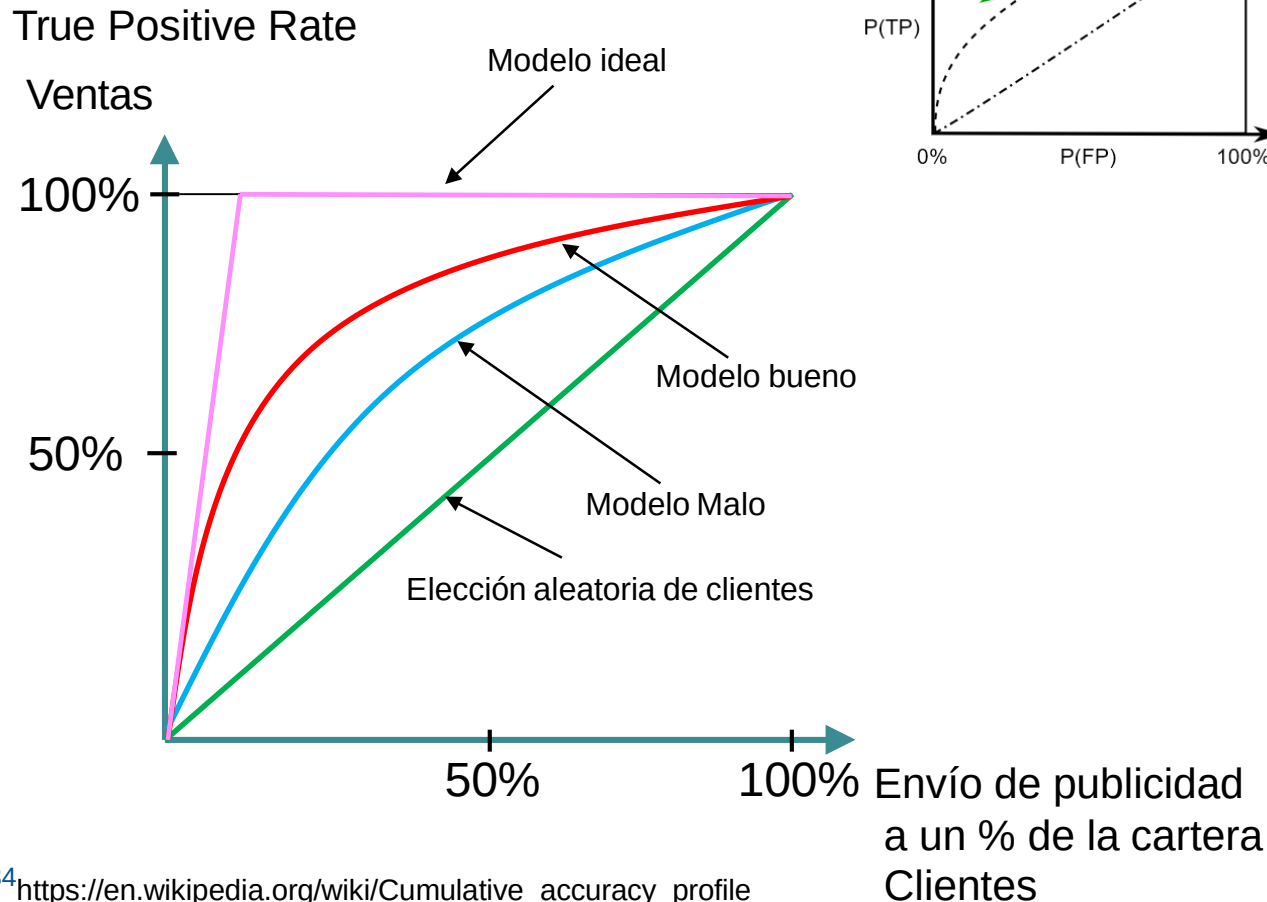
$$\text{Specificity: } TNR = \frac{TN}{TN+FP} = \frac{15}{18} = 0.83$$

https://en.wikipedia.org/wiki/Confusion_matrix

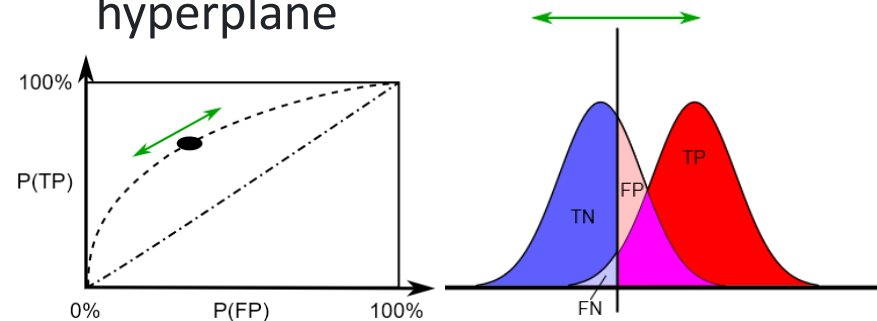
<https://www.juanbarrios.com/matriz-de-confusion-y-sus-metricas/>

Cumulative Accuracy profile (CAP)

La curva representa el porcentaje de ventas acumulado frente al porcentaje de clientes contactados (parámetro de clasificación).



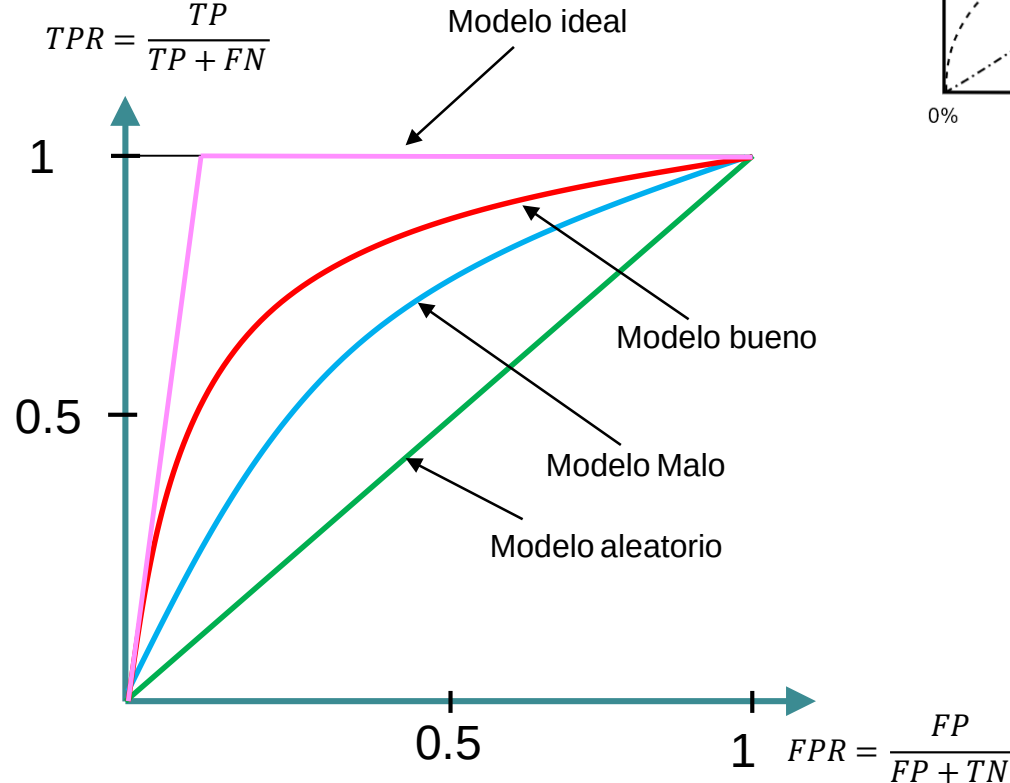
proportional to the distance of the samples X to the separating hyperplane



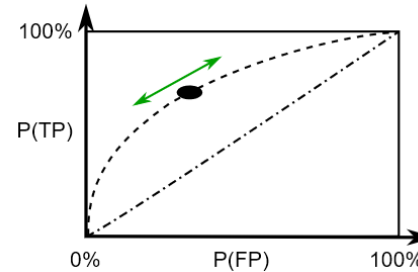
Receiver Operating Characteristic (ROC)

True Positive Rate

$$TPR = \frac{TP}{TP + FN}$$



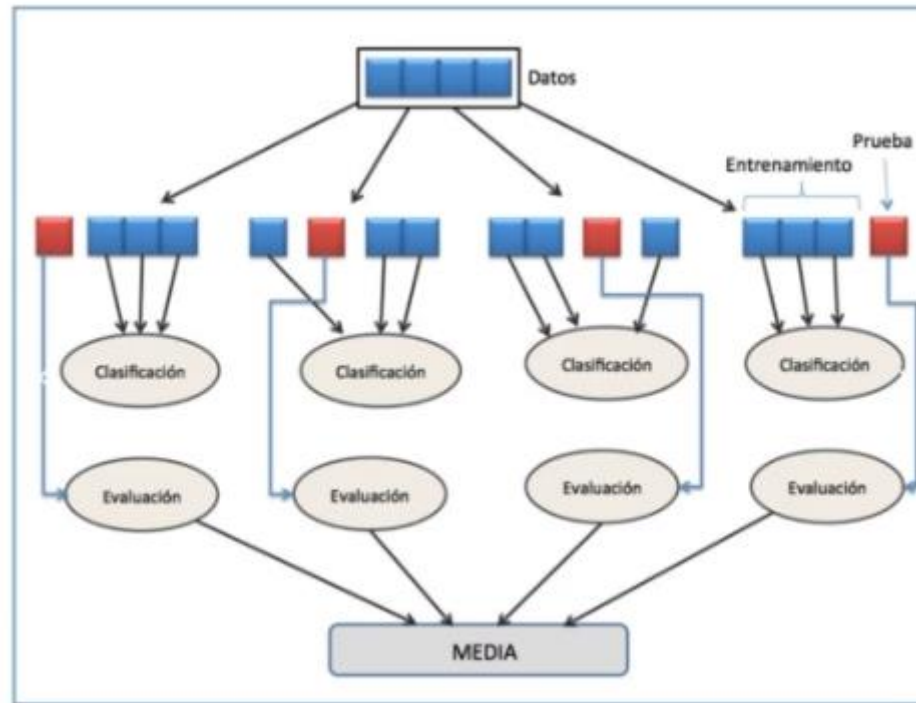
proportional to the distance of the samples X to the separating hyperplane



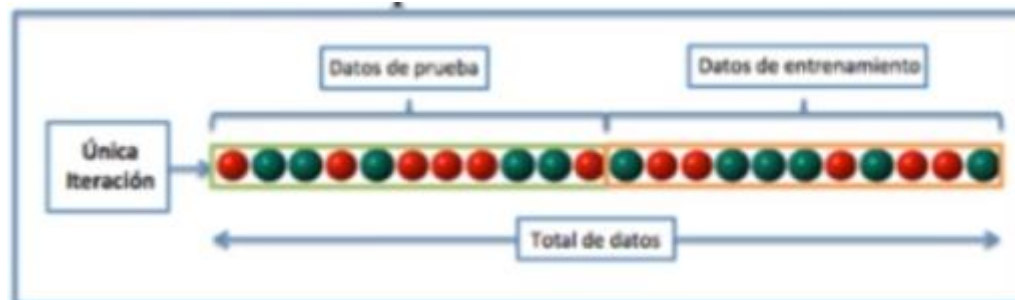
$$TPR = \frac{TP}{TP + FN}$$

Cross Validation

Ejemplo de k-fold Cross Validation con $k = 4$ y 1 clasificador



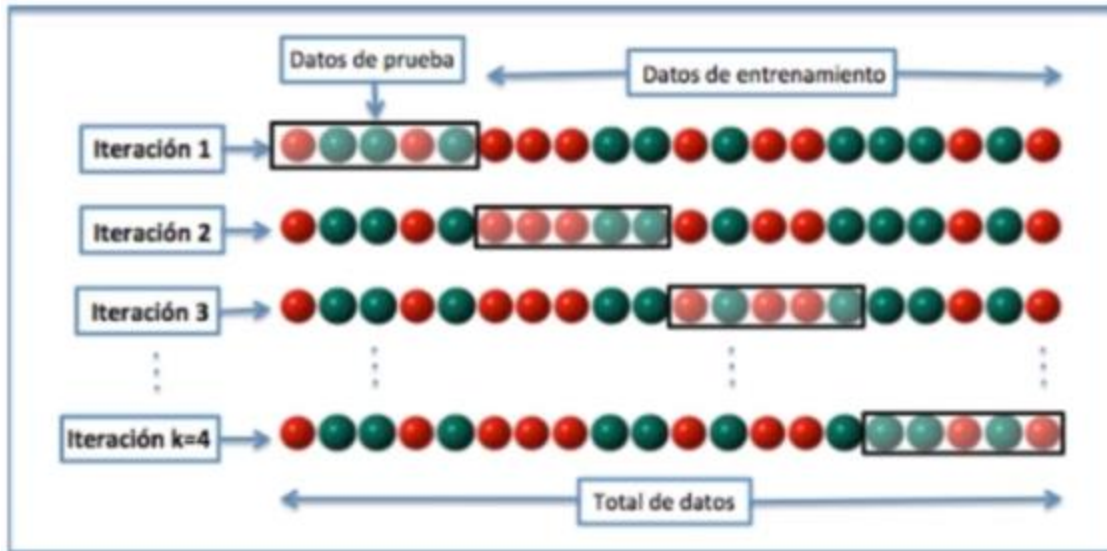
Método de partición de los datos



Cross Validation

Validación cruzada de K iteraciones

K-fold Cross Validation

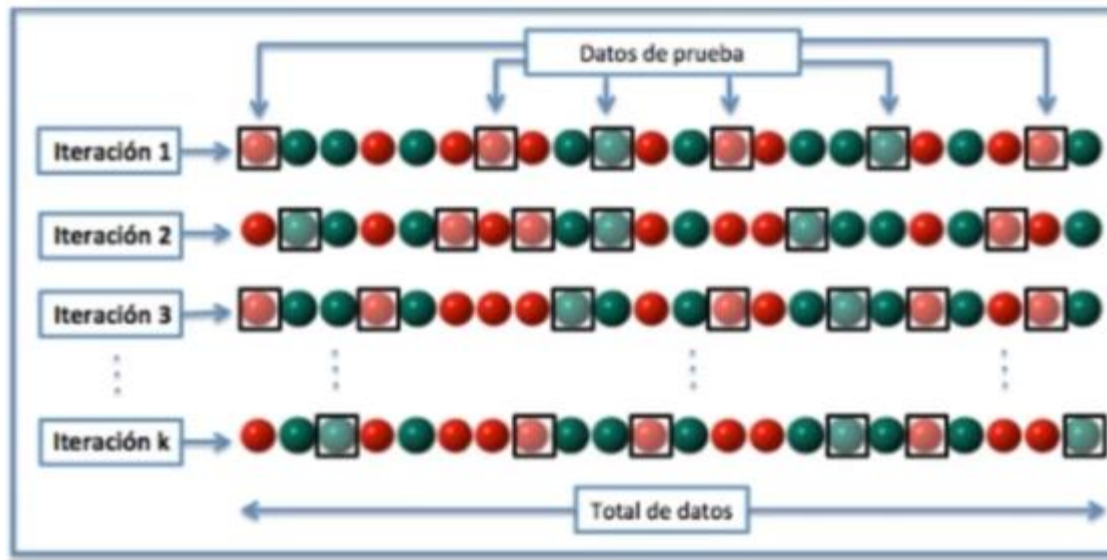


$$E = \frac{1}{K} \sum_{i=1}^K E_i$$

Cross Validation

Validación cruzada aleatoria

Random Cross Validation

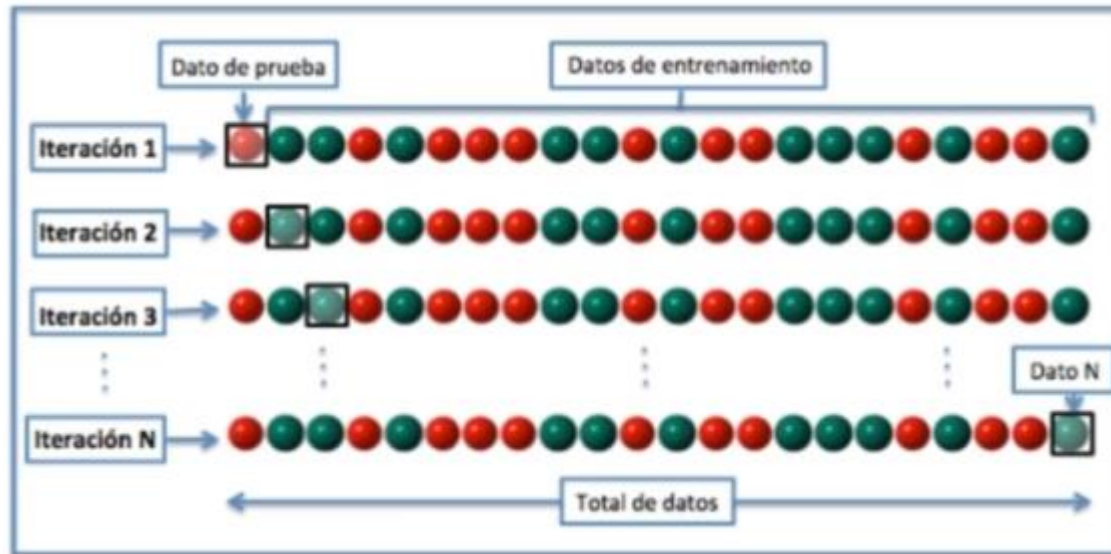


$$E = \frac{1}{K} \sum_{i=1}^K E_i$$

Cross Validation

Validación cruzada dejando uno fuera

Leave one out Cross Validation (LOOCV)



$$E = \frac{1}{N} \sum_{i=1}^N E_i$$

Cross Validation

Ejemplo de k-fold Cross Validation con $k = 4$ y 4 clasificadores

